CHAPTER 8

USE OF STATISTICAL TOOLS

CHAPTER ESSENCE AT A GLANCE:
Integrate statistical techniques into applied economic research: designing an empirical project, collecting primary or secondary data, applying statistical tools, and interpreting findings.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Empirical Economic Project Workflow:** Sequential stages: Statement of the problem, Formulating research hypotheses, Designing data collection schedule, Statistical tabulation, Analytical computations, and Conclusion and Recommendations.

• **Tool Selection Suitability:** Choosing appropriate statistical techniques: Central tendency (mean, median) for summary levels; Dispersion for inequality; Correlation for relationships; Index numbers for trends.

• **Data Ethics and Critical Scrutiny:** Maintaining objective neutrality, avoiding selective cherry-picking of data, acknowledging margins of error, and citing secondary sources.

• **Policy Translation of Research:** Translating statistical insights (e.g. rising unemployment or falling agricultural terms of trade) into concrete economic policy recommendations for planners.

ESSENTIAL ECONOMIC MODELS & GOVERNING FORMULAS

$$\text{Applied Project Pipeline} = \text{Hypothesis} \rightarrow \text{Sample Survey} \rightarrow \text{Computation} \rightarrow \text{Policy Synthesis}$$

Empirical Research Methodological Cycle

$$\text{Tool Selection Matrix} = \text{Averages (Level)} + \text{Correlation (Link)} + \text{Index (Trend)}$$

Statistical Analytical Tool Selection Strategy

$$\text{Sample Proportion Estimate } (p) = \frac{\text{Number of Favourable Cases } x}{\text{Total Sample Size } n}$$

Empirical Survey Attribute Prevalence Rate

$$\text{Empirical Reliability} \iff \text{Objective Sampling} + \text{Rigorous Verification} + \text{Transparent Margins}$$

Scientific Economic Research Integrity Standard

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Statistical analysis cannot prove an economic hypothesis with 100 percent certainty; it only provides empirical support within statistical confidence bounds.

• **Alert:** Never use complex statistical formulas without verifying basic data assumptions like normality or linearity.

• **Alert:** Always state the sample size and data collection limitations clearly in the project conclusion.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State the first two sequential steps in conducting an empirical statistical project in economics. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Identifying the Problem):** Clearly defining the specific economic research question (e.g. impact of petrol price rise on household budgets). **[1 Mark]**

• **Step 2 (Formulating Hypotheses):** Setting testable propositions regarding expected economic relationships. **[1 Mark]**

2. State two ethical standards that must be observed while collecting and analyzing statistical data. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Objectivity:** Recording actual collected observations without personal subjective tampering or fabricating figures. **[1 Mark]**

• **Confidentiality:** Protecting personal and financial privacy of survey respondents without unauthorized public disclosure. **[1 Mark]**

3. Explain how different statistical tools (Mean, Correlation, and Index Numbers) are integrated to analyze a socio-economic problem. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Arithmetic Mean:** Establishes typical baseline levels (e.g. average monthly family expenditure). **[1 Mark]**

• **Correlation Analysis:** Examines whether rising income is statistically linked with changes in food versus education expenditure. **[1 Mark]**

• **Index Numbers:** Measures temporal trends showing how living costs have changed relative to the base year. **[1 Mark]**

4. A student wants to study 'Consumer Expenditure on Fast Food Among Teenagers'. Suggest: (a) Suitable method of data collection, (b) Sample size, (c) Two key questions for the questionnaire. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **(a) Collection Method:** Direct personal questionnaire or online survey form administered at school or college campuses. **[1 Mark]**

• **(b) Sample Size:** Representative sample of 100 to 150 students across different age and income groups. **[1 Mark]**

• **(c) Two Questions:** (1) 'How many times per week do you consume fast food?', (2) 'What is your average weekly expenditure on fast food?'. **[1 Mark]**

5. State the essential elements that should be included in the Final Report of a statistical research project. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Title and Objectives:** Statement of the research topic and primary aims of investigation. **[1 Mark]**

• **Methodology and Data Tables:** Sampling design, questionnaire copy, and classified statistical frequency tables and charts. **[1 Mark]**

• **Findings and Policy Suggestions:** Summary of empirical findings, limitations, and actionable policy recommendations. **[1 Mark]**

6. Why is it important to state the limitations of data in a statistical project report? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Preventing Misleading Conclusions:** Alerting readers and policymakers about potential sampling or non-sampling error margins. **[1.5 Marks]**

• **Academic Integrity:** Demonstrating scientific rigor and contextualizing results to the surveyed sample universe. **[1.5 Marks]**

7. Explain the major stages in executing a complete Statistical Project in Economics from Problem Identification to Drafting the Report. [Score: 4 Marks]

Answer & Key Valuation Steps:

- **Stage 1 & 2:** Identifying the economic problem and formulating clear research objectives and hypotheses. **[1 Mark]**
- **Stage 3 & 4:** Designing the questionnaire schedule and conducting data collection via representative sampling. **[1 Mark]**
- **Stage 5 & 6:** Organizing raw responses into tables and computing statistical measures (averages, correlation, percentages). **[1 Mark]**
- **Stage 7:** Interpreting statistical findings and writing the final project report with policy recommendations. **[1 Mark]**